

ICAIL – Round 2 – Group No. 3

Project Proposal

Project Title

----- **SMS Spam Detection** -----

	Project No.: 6	
	<i>Name:</i>	<i>ID.</i>
1	Ahmed Salah Ebrahim	2025-icail-1183
2	Walaa Fathy	2025-icail-1180
3	Mohamed Hamza	2025-icail-1194
4	Nehal Sameh	2025-icail-1165
5	Abdelrahman Mohamed Atef	2025-icail-1115

1. Introduction

Nowadays, mobile phones are at the center of everything we do. We use them to make calls, check messages, manage work, and stay connected with others. They're always in our hands-making them the easiest and fastest way to communicate. But with this convenience comes a risk.

Every day, users receive SMS messages that may look normal but are actually dangerous. These can include fake messages from banks asking for personal data, links to suspicious websites promising rewards, or even scams pretending to be someone you know. Many people fall for these tricks without knowing it.

That's why this project is important. We are building a smart system that uses Artificial Intelligence (AI) to detect and stop spam messages before they harm users. The system will automatically scan SMS messages, identify those that look suspicious or dangerous, move them away from your inbox, and warn you so you can stay safe.

2. Project Objectives

This project has several clear goals:

- Protect users from scam or harmful SMS messages using smart detection.
- Automatically tell the difference between real messages and spam.
- Raise awareness about SMS threats and help users recognize dangerous messages.
- Show that AI can be used to defend users and make scammers' efforts useless.
- Deliver a simple, easy-to-use solution that requires no technical background.

3. Target Users

Our target users include:

- Everyday mobile phone users who want to protect their personal data and stay safe.
- Business professionals and employees who handle sensitive information through mobile communication.
- Companies and organizations that want to protect their staff and data from SMS scams.

In simple words, this tool is for anyone who values their security, privacy, and peace of mind while using their phone.

4. Key Features

The core features of our SMS Spam Detection model will include:

- **Spam Detection:** Accurately detect and classify spam messages in real-time.
- **Automatic Filtering:** Move spam messages to a separate folder to keep your inbox clean.
- **User Warning System:** Display a warning message when a suspicious SMS is received.
- **Curiosity Control:** Prevent users from accidentally opening dangerous messages out of curiosity.

5. Tools and Technologies

To develop and train our AI model, we will use:

- **Language:** Python – easy to use, powerful, and widely supported in AI.
- **Platform:** Google Colab – a cloud-based tool that helps us write and test code online with free access to GPU.
- **Libraries and Frameworks:**
 - Pandas – for reading and organizing the dataset.
 - NumPy – for numerical calculations.
 - Scikit-learn – for training machine learning models like Naive Bayes or Logistic Regression.
 - TensorFlow & Keras – for building deep learning models if needed.
- **Dataset:** We will use the SMS Spam Collection Dataset, a public dataset containing thousands of labeled messages as “spam” or “ham” (not spam).

6. Expected Outcome

By the end of the project, we expect to deliver:

- A working spam detection system that automatically scans and filters SMS messages.
- A simple demo showing how the system works with real examples.

Ultimately, we aim to develop a reliable AI-based tool that enhances user safety and awareness, ensuring that mobile users can confidently interact with their SMS inbox without fear of manipulation or fraud.